

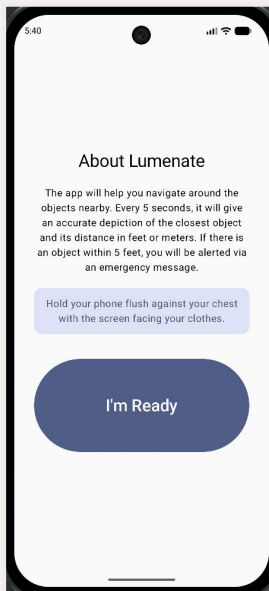
Luminate *Update* 2

MONTH YEAR

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Speech Recognition



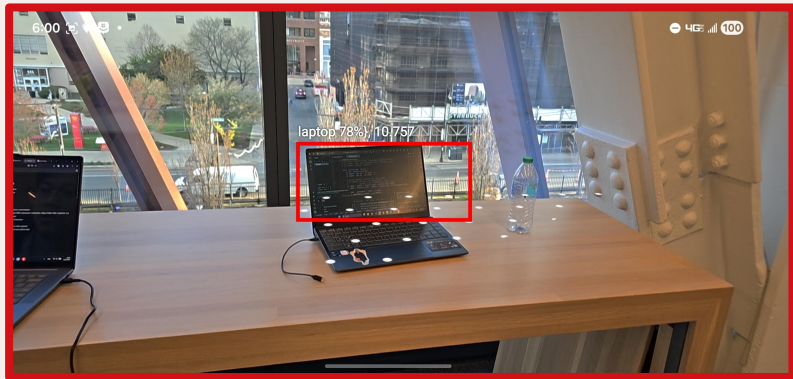
Using Android's built in `SpeechRecognizer` library, we were able to construct a coroutine that allows users to navigate through screens using just their voice. While facing this screen, if the user says the word "ready" in any capacity the app will open the camera screen.

In the future, we will add voice accessibility for navigating all screens, changing preferences, and allowing permissions.

```
631 private fun continuousSpeechFlow(context: Context): Flow<String> = callbackFlow {
632     val audioManager = context.getSystemService<AudioManager>(Context.AUDIO_SERVICE) as AudioManager
633     if (!SpeechRecognizer.isRecognitionAvailable(context)) { close(); return@callbackFlow }
634     val recognizer = SpeechRecognizer.createSpeechRecognizer(context)
635
636     fun restart() {
637         audioManager.muteBeep()
638         recognizer.startListening(recognizerIntent = sttIntent)
639     }
640
641     recognizer.setRecognitionListener(object : RecognitionListener {
642         override fun onReadyForSpeech(params: Bundle?) {}
643         override fun onBeginningOfSpeech() {}
644         override fun onEndOfSpeech() {}
645         override fun onBufferReceived(buffer: ByteArray?) {}
646         override fun onEndOfSpeech() {}
647         override fun onPartialResults(partialResults: Bundle?) {}
648         override fun onEvent(eventType: Int, params: Bundle?) {}
649         override fun onResults(results: Bundle?) {
650             val transcript = results.getStringArrayList(key = SpeechRecognizer.RESULTS_RECOGNITION)?.firstOrNull() ?: ""
651             if (transcript.isNotBlank()) trySend(element = transcript)
652             restart()
653         }
654     })
```

Custom SpeechRecognizer

Notification System

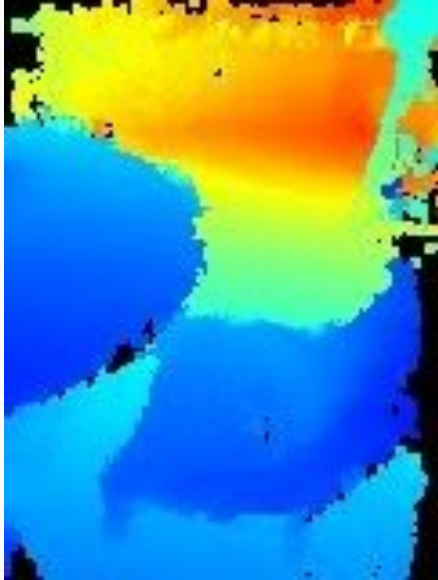


Additionally, the TTS system will be overhauled to use Google's voices through their API. We will have to use a MediaPlayer object which requires the requesting of audio focus to ensure that no other background apps are playing audio while Lumenate reads what it needs to.

The app consistently alerts the user of up to 5 detected objects and their distances, with the next set of objects read aloud 5 seconds after the previous set of objects has been fully read.

We will implement emergency alerts to notify the user of any object that comes within 8 feet of them.

Raw Depth API



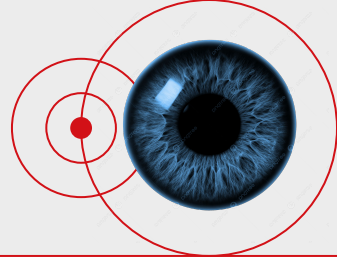
Raw Depth API is implemented, Lumenate can now detect distance of the objects detected from the EfficientDet model from the mobile device

Every time the app gets an object detected, the API will grab that object and measure the distance/depth of it

That distance gets spit back out via TTS

If there is no known distance, it outputs unknown

Debugging & Testing Strategies



Emulation limitations:

- Depth API does not work on emulators → redirects the emulating phone to google play
- Lack of real-life mobility to walk around and use the app
- Microphone speech-to-text non-functional
 - MP3 file upload needed to test

Solution:

- Jimmy and Ilay tested using Android hardware (phones)

QA:

- Break points on logging (Log.d) for distance calculations
- Break points on logging (Log.d) for object detection
- Comparing running performance to user flow diagram we first presented

Core Features To Be Continued

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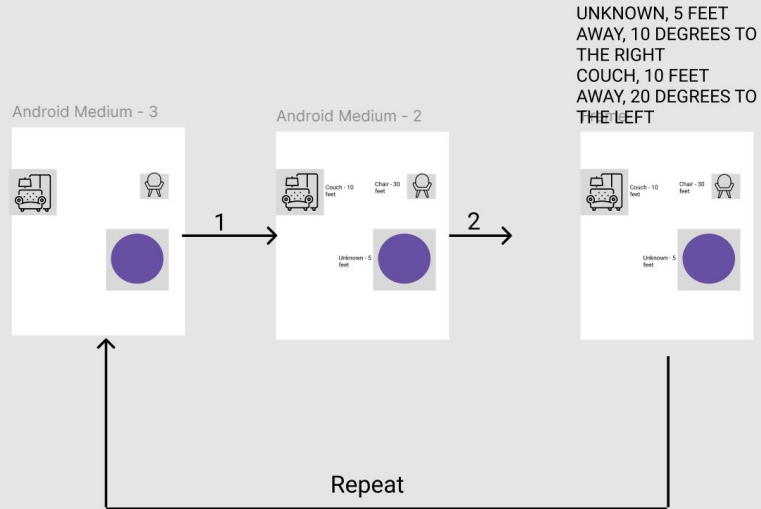
Feasible features to be added for final update:

- Speech commands
- Extra TTS voices - Google API
 - Gets persistently stored
- Improved distance readings
- Enrichment of distance readings using relative bearings
(i.e. "This chair is *x degrees*, 3 meters to your left)
- Settings persistence

Descoped: Astar navigation to safely direct users to an object in the room

Navigation Flow

<https://www.figma.com/design/IFBztQrkyx68UfyB0R52k9/project-update-1?node-id=1155174-791&p=f&t=iT1BHYUraHx3RzCA-0>



Thank you